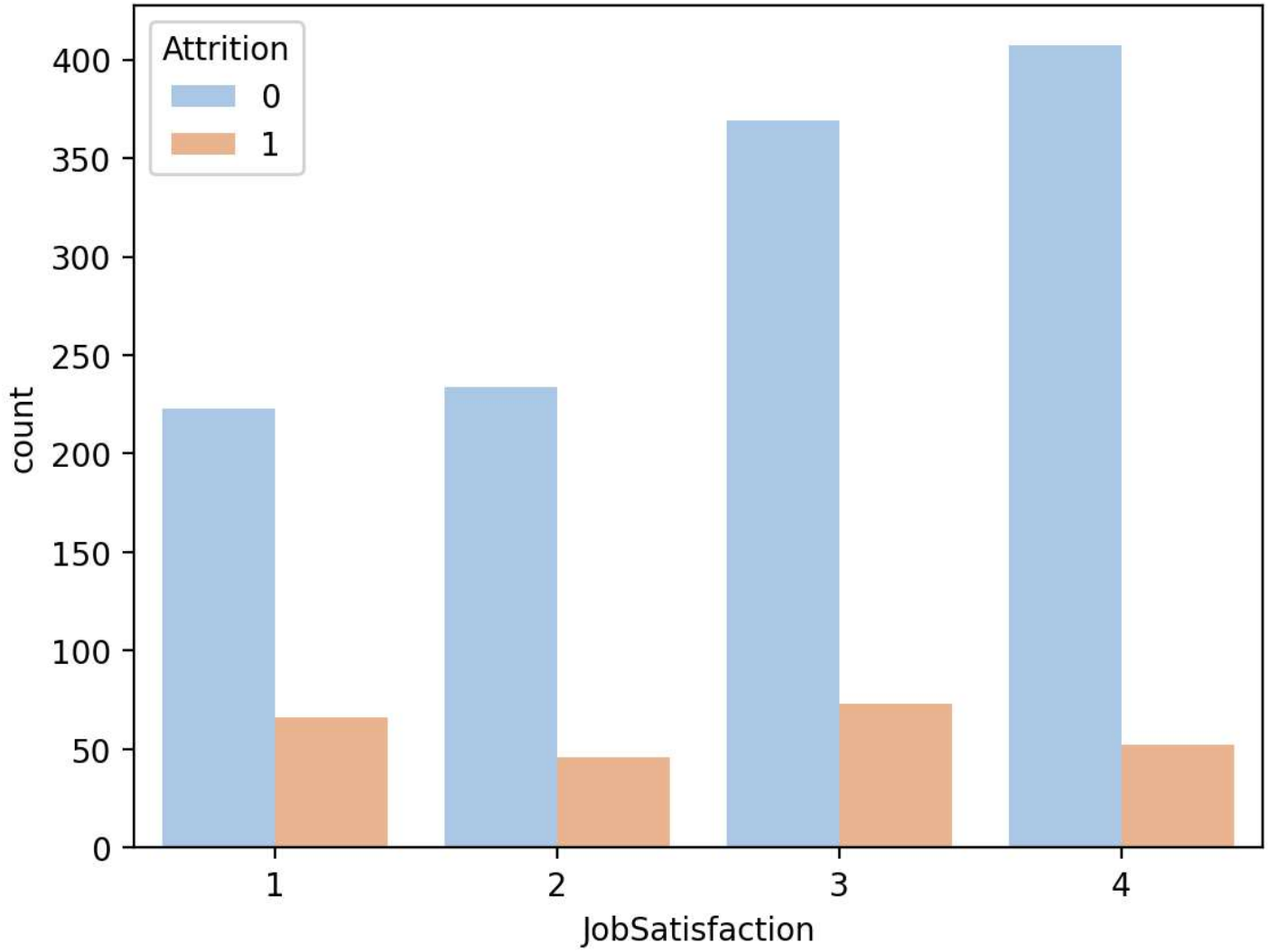
📊 Employee Attrition Analysis & Prediction Dashboard

HR Analytics for Smart Retention Strategy

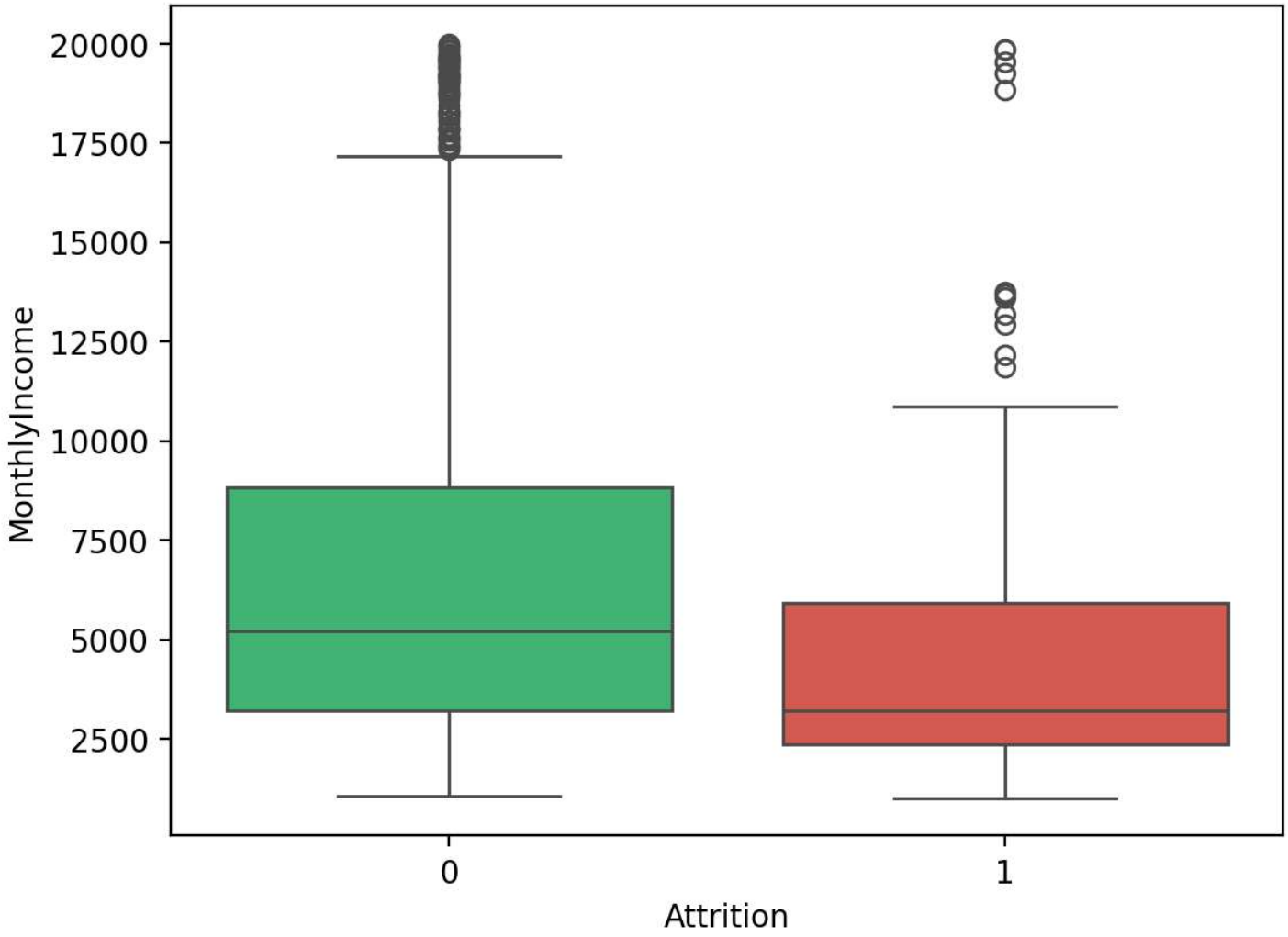
📌 Key Performance Indicators

👥 Total Employees	🟧 Employees Left	📉 Attrition Rate (%)
1470	237	16.12

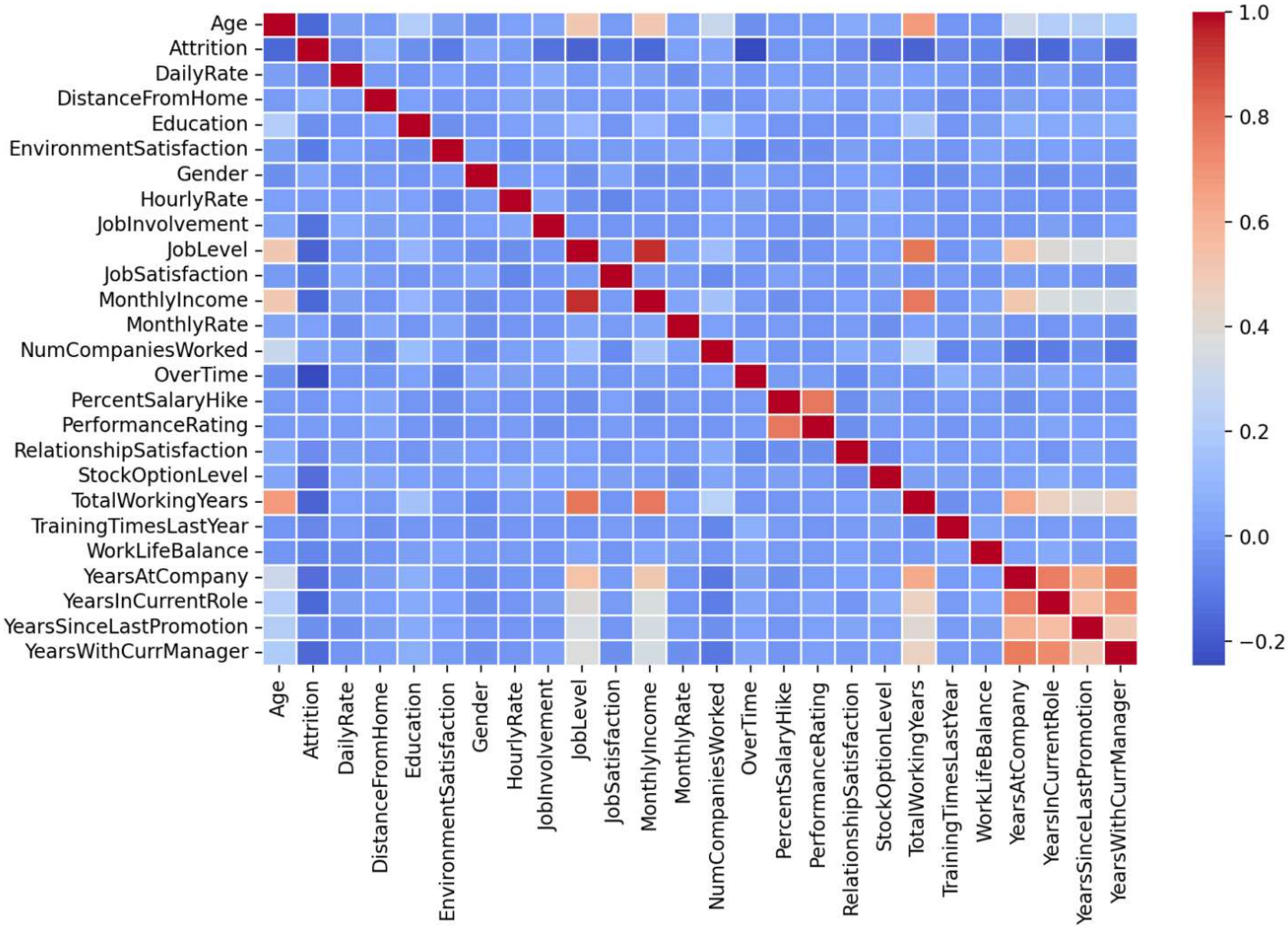
📊 Attrition by Job Satisfaction



💰 Salary Distribution vs Attrition



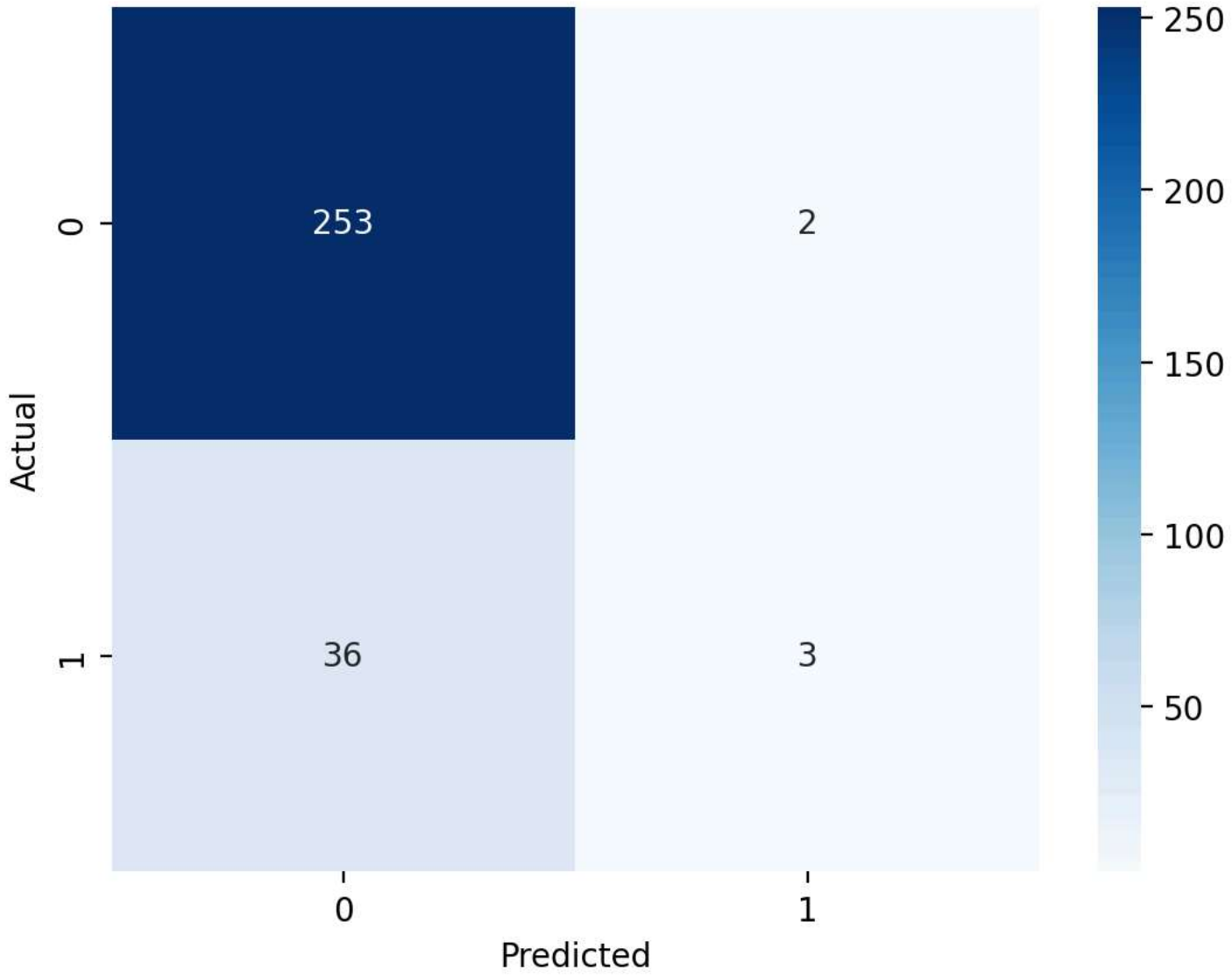
📈 Correlation Heatmap



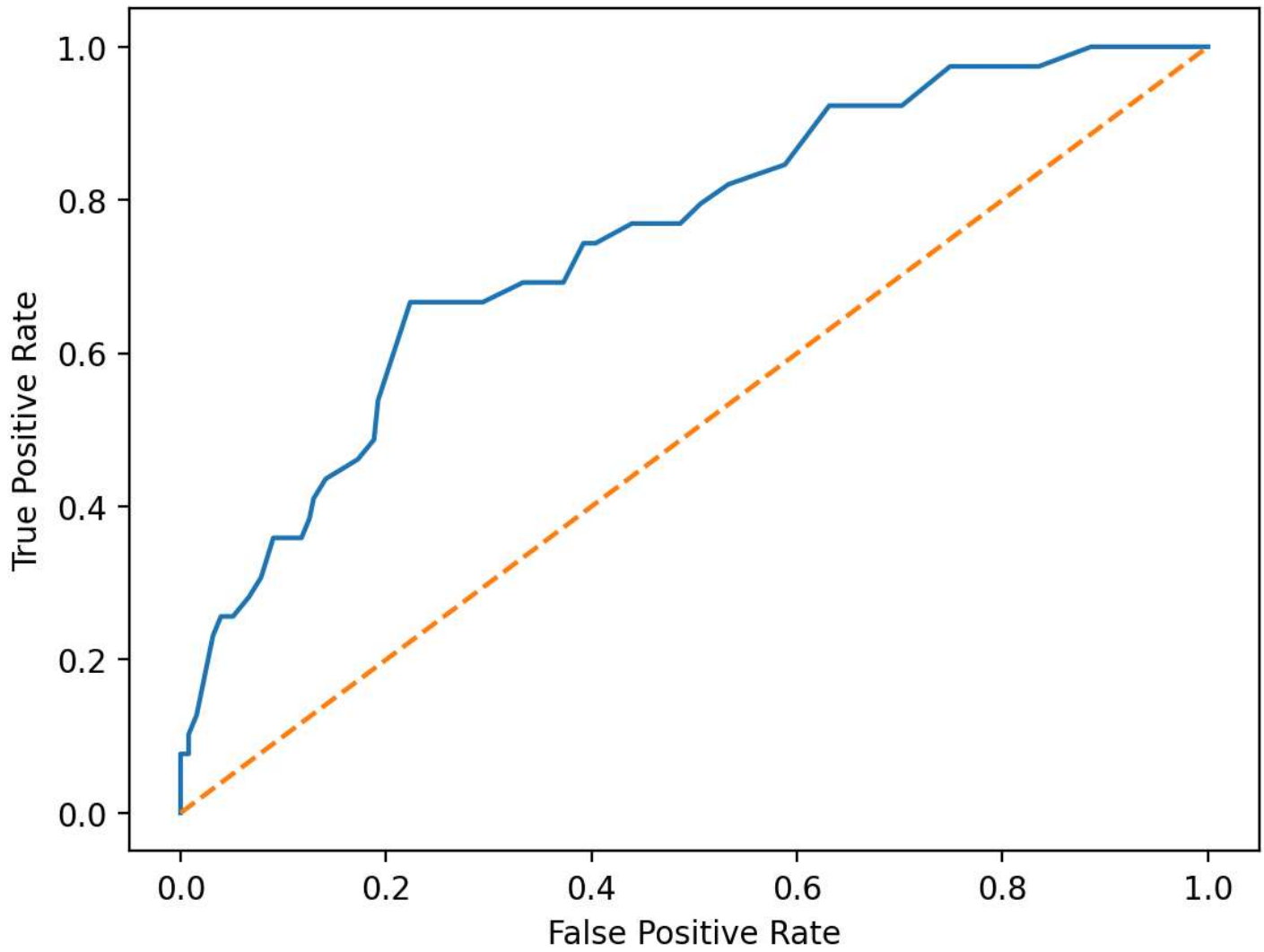
🤖 Model Training & Evaluation

Accuracy	Precision	Recall
0.871	0.6	0.077
F1 Score	**AUC-ROC**	**Training Time (sec)**
0.136	0.749	0.359

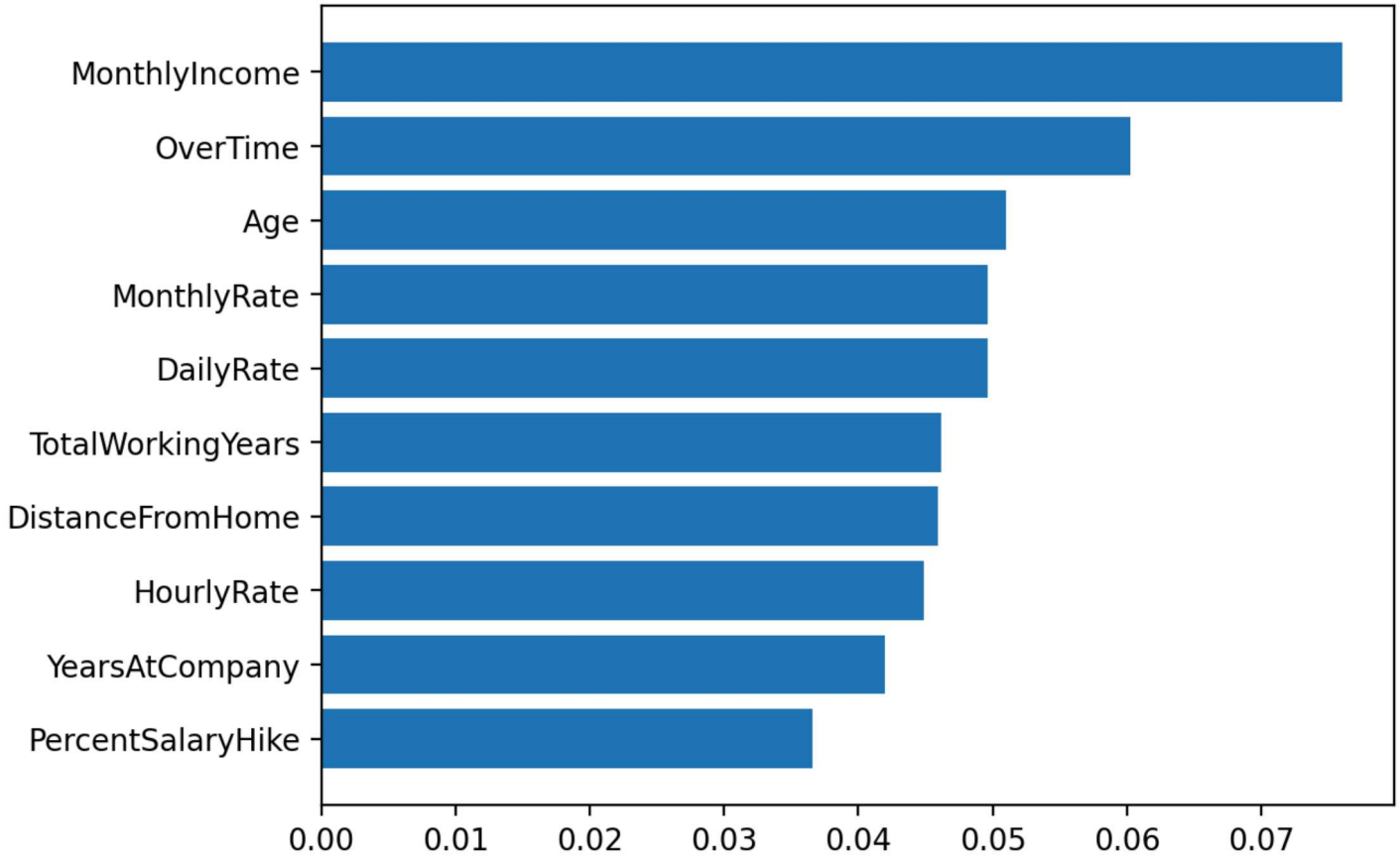
📊 Confusion Matrix



📈 ROC Curve



📌 Top 10 Important Features



🚨 High Risk Employees (Probability > 70%)

	Age	Attrition	DailyRate	DistanceFromHome	Education	EnvironmentSatisfaction	Gender	HourlyRate	JobInvolvement	JobLevel	JobSatisfaction	MonthlyI
0	41	1	1102	1	2	2	0	94	3	2	4	
2	37	1	1373	2	2	4	1	92	2	1	3	
14	28	1	103	24	3	3	1	50	2	1	3	
21	36	1	1218	9	4	3	1	82	2	1	1	
26	32	1	1125	16	1	2	0	72	1	1	1	
33	39	1	895	5	3	4	1	56	3	2	4	
34	24	1	813	1	3	2	1	61	3	1	4	
36	50	1	869	3	2	1	1	86	2	1	3	
42	26	1	1357	25	3	1	1	48	1	1	3	
69	36	1	318	9	3	4	1	79	2	1	3	

💼 Business Impact Analysis

📌 Attrition Rate Reduction

Identifies high-risk employees early and enables:

- Targeted retention programs
- Compensation adjustments
- Career growth planning
- Work-life balance improvements

💰 Cost Savings

Reduces:

- Recruitment costs
- Onboarding expenses
- Training investment losses

📊 Strategic HR Decisions

Transforms HR from reactive hiring to proactive retention strategy.

Developed as an HR Analytics Machine Learning Project